Year II — Mistake & Methods Log

Based on lectures by Various Lecturers

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Contents

1 Mathematical Methods II **2**
1.1 Line Integrals . . . 2
1.1.1 Mistakes and Tips . . . 2
1.1.2 Methods . . . 2
1.2 Surface and Volume Integrals . . . 3
1.2.1 Mistakes and Tips . . . 4

2 Algebra II **5**
2.1 Rings, Fields and Subrings . . . 5
2.1.1 Mistakes and Tips . . . 5
2.1.2 Methods . . . 6

Chapter 1

Mathematical Methods II

1.1 Line Integrals

1.1.1 Mistakes and Tips

What actualy is $f(\mathbf{x}(t))$?

$f(\mathbf{x}(t))$ is a function that acts on the entries for $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))$ of a curve.

| $\frac{dx}{dt}$ | differs

| $\frac{d\mathbf{x}}{dt}$ | is not always a fixed number and may be a function of t that differs.

Parametrising implicit curves

Sometimes you need to parametrise the function before you can integrate under the scalar field. Make sure to remember your polar coordinates and draw a diagran if possible

Splitting piecewise smooth curves

When a curve is piecwise smooth, make sure to split it when integrating.

Unit tangents and orthonormal

You can always find the unit tangent vectors to a curve but they are not always orthonormal

'Length of a curve'

If it asks for the length of a curve, use arc length

1.1.2 Methods

Method 1.1.1 (Evaluating a Scalar Line Integral)**.**

(i) Write $f(\mathbf{x}), \mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))$ and $t \in [t_0, t_1]$

(ii) Find $\frac{d\mathbf{x}}{dt}$ and then take the magnitude (this may not be an integer but a function of t)

(iii) Substitute in and integrate

Method 1.1.2 (Evaluating arc length)**.**

(i) Write $\mathbf{x}(t) = (x_1(t), x_2(t), x_3(t))$

(ii) Find $\frac{d\mathbf{x}}{dt}$

(iii) Perform $\sqrt{\frac{d\mathbf{x}}{dt} \cdot \frac{d\mathbf{x}}{dt}}$

(iv) Use the formula for arclength

Method 1.1.3 (Evaluating a Vector Field Line Integral)**.**

(i) Sketch $\mathbf{x}(t)$ if possible

(ii) Substitute $\mathbf{x}(t)$ into $\mathbf{F}(\mathbf{x})$ to form $\mathbf{F}(\mathbf{x}(t))$

(iii) Find $\frac{d\mathbf{x}}{dt}$

(iv) Perform $\mathbf{F}(\mathbf{x}(t)) \cdot \frac{d\mathbf{x}}{dt}$

(v) Substitute into the formula

Method 1.1.4 (Finding unit tangent vectors)**.** To find the unit tangent vectors for the coordinate system $\mathbf{x}(u, v, w)$ use the formulas:

$$\boldsymbol{e}_u = \frac{1}{h_u}\frac{\partial \boldsymbol{x}}{\partial u}, \quad \boldsymbol{e}_v = \frac{1}{h_v}\frac{\partial \boldsymbol{x}}{\partial v}, \quad \boldsymbol{e}_w = \frac{1}{h_w}\frac{\partial \boldsymbol{x}}{\partial w},$$

where

$$h_u = \left|\frac{\partial \boldsymbol{x}}{\partial u}\right|, \quad h_v = \left|\frac{\partial \boldsymbol{x}}{\partial v}\right|, \quad h_w = \left|\frac{\partial \boldsymbol{x}}{\partial w}\right|$$

are called the scale factors.

Method 1.1.5 (Ensuring an Orthonormal Basis)**.** To check if these unit tangent vectors form an orthonormal basis we will have to ensure that they are all mutually orthogonal.

Method 1.1.6 (Finding components of field in different coordinate system)**.**

- Find f_x, f_y, f_z and re-express them in the coordinate system for $\mathbf{x}(u, v, w)$ using the conversions. f_x,$_y$, f_z refers to the functions / components along the corresponding axes in this context.
- Find e_u, e_v, e_w using either memorisation or derivation from the formula.
- Follow the general pattern that $f_v = e_v \cdot$ to find all the components.t

1.2 Surface and Volume Integrals

1.2.1 Mistakes and Tips

Surface Area vs Surface Integral in a Scalar Field

Surface area will contribute equal weighting from all positions in a scalar field (this being 1). When finding our surface area, this means we take the integral to be:

$$\mid S \mid = \int_S 1 \, dS = \int_V \mid \frac{\partial \mathbf{x}}{\partial u} \times \frac{\partial \mathbf{x}}{\partial v} \mid \, du \, dv$$

When we are operating in a scalar field, each point may have a value other than 1 given to it by a scalar function. To accumulate all of the weights, we integrate via the function:

$$\mid S \mid = \int_S f(\mathbf{x}(u,v)) \; dS = \int_V f(\mathbf{x}(u,v)) \mid \frac{\partial \mathbf{x}}{\partial u} \frac{\partial \mathbf{x}}{\partial v} \mid \, du \, dv$$

Black Box

Volume will contribute equal weighting from all positions in a scalar field (this being 1). When finding our surface area, this means we take the integral to be:

$$\mid V \mid = \int_S 1 \, dS = \int_V \mid \frac{\partial \mathbf{x}}{\partial u} \frac{\partial \mathbf{x}}{\partial v} \mid \, du \, dv$$

When we are operating in a scalar field, each point may have a value other than 1 given to it by a scalar function. To accumulate all of the weights, we integrate via the function:

$$\mid S \mid = \int_S f(\mathbf{x}(u,v)) \; dS = \int_V f(\mathbf{x}(u,v)) \mid \frac{\partial \mathbf{x}}{\partial u} \frac{\partial \mathbf{x}}{\partial v} \mid \, du \, dv$$

Chapter 2

Algebra II

2.1 Rings, Fields and Subrings

2.1.1 Mistakes and Tips

Determining unknown binary operations

If not specified, assume that R is a ring with $+$ and $\cdot$ by default

Using universal conditions

Work with any universal condition when manipulating or showing structure and then use strategically chosen elements

Strategically choosing elements

Use strategically chosen elements that exploit certain conditions such as $a+b$ or ab

Dividing in rings

Never divide by elements unless you know it is invertible. This is because rings do not require multiplicative inverses. Instead use addition / subtraction on either side and also distributivity.

Make use of earlier results

Utilise earlier results of the question where possible.

Simplifying expansions

Do not simplify expansions with hypotheses until it is the direct next step as you lose the algebra needed to show it is true.

Utilising LHS and RHS

Showing properties like associativity, commutativity and distributivity can be done easily by showing LHS = RHS.

Efficiently forming Modular Arithmetic tables

When dealing with integers modulo n, it is smart to use a multiplication and addition table. To quickly fill them in, consider how much over a multiple of n your entry to the table is to find the residue.

Using fields for subrings

Using the fact that a subring of a ring is a ring itself allows you to quickly prove rings if you know that they are a subring of a larger field like $\mathbb{C}$.

Dealing with subring questions

When dealing with a subring question, make sure to use inherited elements and operations to show certain structures. For example 0 and 1.

Showing subrings are the parent rings

Showing subrings are the ring themselves requires showing $R \subset S$ as $S \subset R$ by default.

Using counterexamples for disproofs

Finding one failure allows you to disprove closure and the full ring quite easily. Use counterexamples strategically here.

2.1.2 Methods

Method 2.1.1 (Determining a Ring)**.** To determine a ring, one should first check that the ring forms an abelian group with respect to the 'addition' operation and that the following requirements hold for the 'multiplication' operation:

i) **Existence of an Identity** - There exists an element $1 \in R$ such that $1 \cdot x = x \cdot 1 = x$, for all $x \in R$.

ii) **Associativity** - For any $x, y, z \in R$ we have $x(yz) = (xy)z$.

iii) **Distributivity** - For any $x, y, z \in R$ we have $x(y + z) = xy + xz$ and $(y + z)x = yx + zx$.

Method 2.1.2 (Showing a Subring)**.** We can quickly show a subring by showing:

- the additive and multiplicative identities exist in the set,
- it's closed under addition
- it's closed under multiplication
- it contains additive inverses.

Method 2.1.3 (Determining a Ring quickly)**.** Rather than displaying a set satisfies all of the properties of a ring, it is sufficient to show that the set is a subring of a ring. It is therefore a ring itself.

Method 2.1.4 (Showing commutativity)**.** Showing commutativity requires introducing arbitrary elements and comparing expressions such as $(a+b)^2$ and the additive inverses to show equivalence.

Method 2.1.5 (Quick way to see that a set is not a ring)**.** If multiplication is not closed by default then a set fails to be a ring as it is a requirement of the binary operation.

Method 2.1.6 (Disproving $\mathbb{Z}/n$ as a field)**.** Showing that $\mathbb{Z}/n$ is not a field requires finding 2 single none zero elements that are zero divisors such that:

$$ab = 0 \bmod(n).$$

This immediately disproves a field. Check columns of a table here to do this quickly.